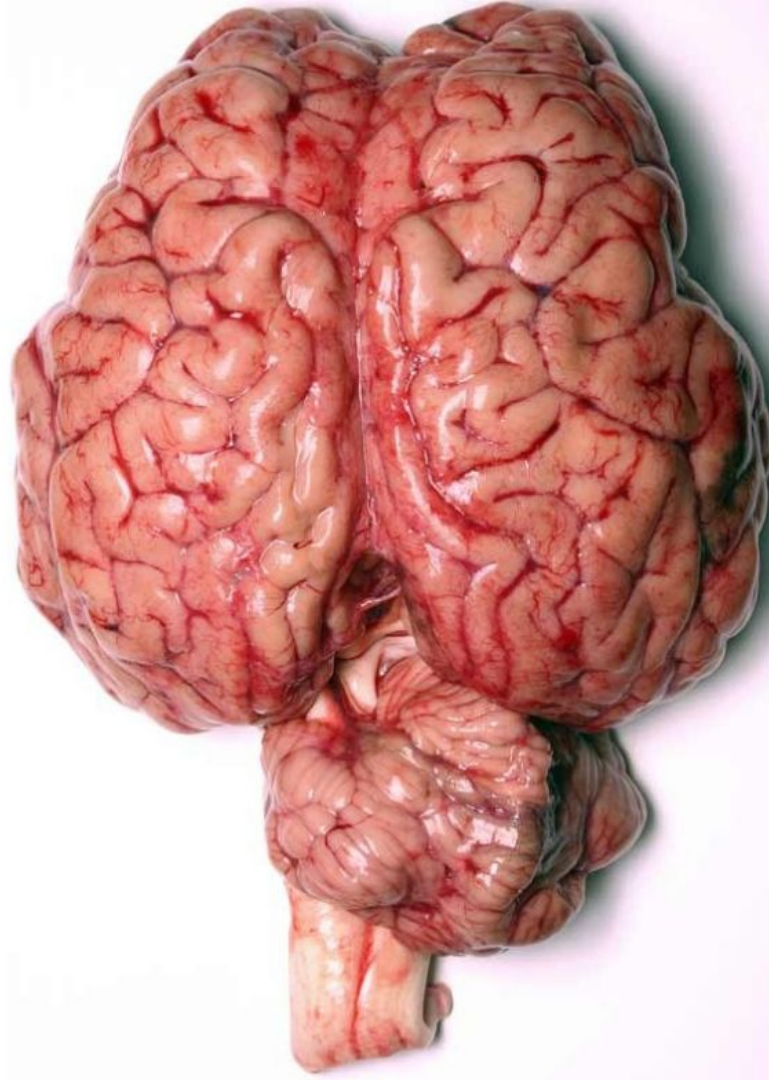


Perceptual consciousness

Nathan Faivre
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<https://nfaivre.netlify.com>



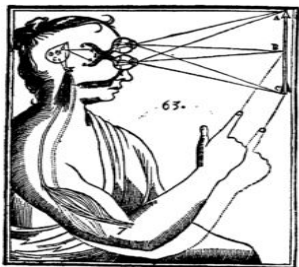




Philosophical zombies (Chalmers, 1996)

The hard problem of consciousness

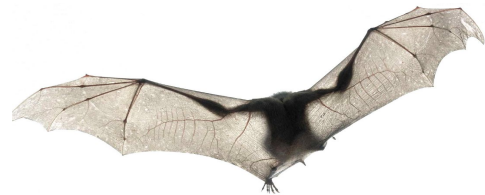
- *The most important problem in the biological sciences* (Searle, 1997)
- *The last surviving mystery* (Dennett, 1991)
- *The most baffling problems in the science of mind* (Chalmers, 1995)
- *With consciousness it seems hopeless* (Nagel, 1974)
- *Nobody has the slightest idea how anything material could be conscious. Nobody even knows what it would be like to have the slightest idea how anything material could be conscious* (Fodor, 1992)



Mind/Body (Descartes)



Fonctions (T. Huxley)



Intimacy and subjectivity (T. Nagel)

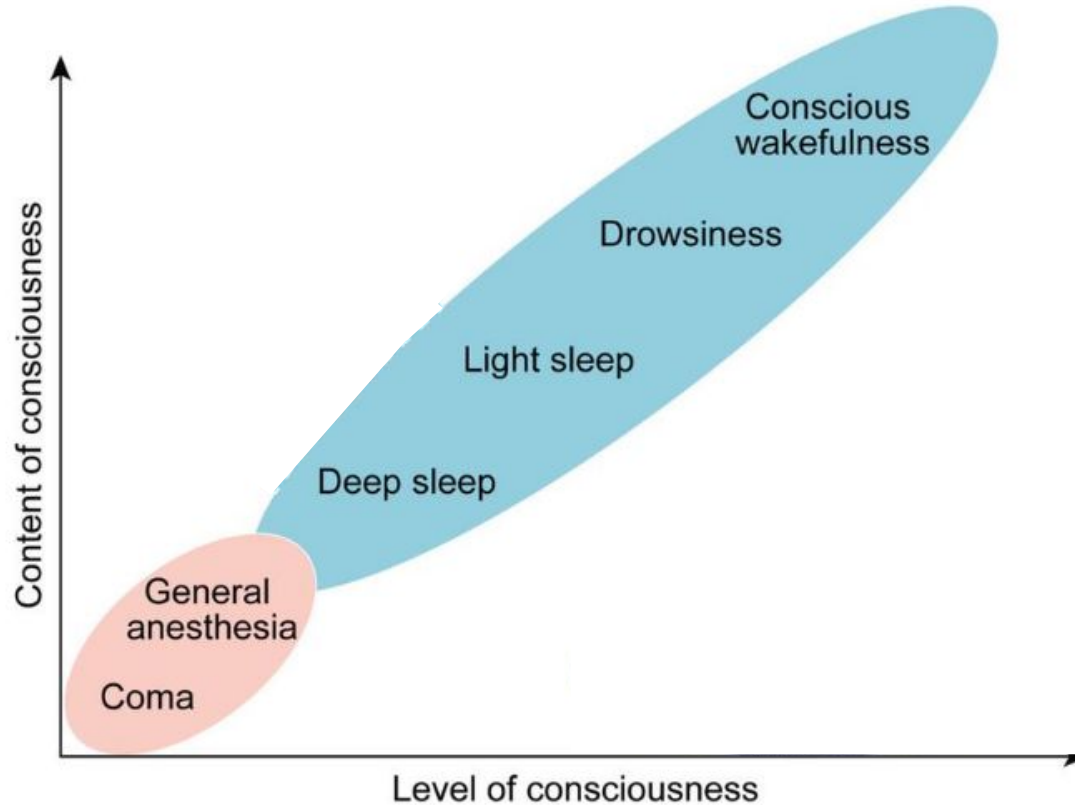
How to define consciousness?

To study consciousness scientifically, one has to define it first.



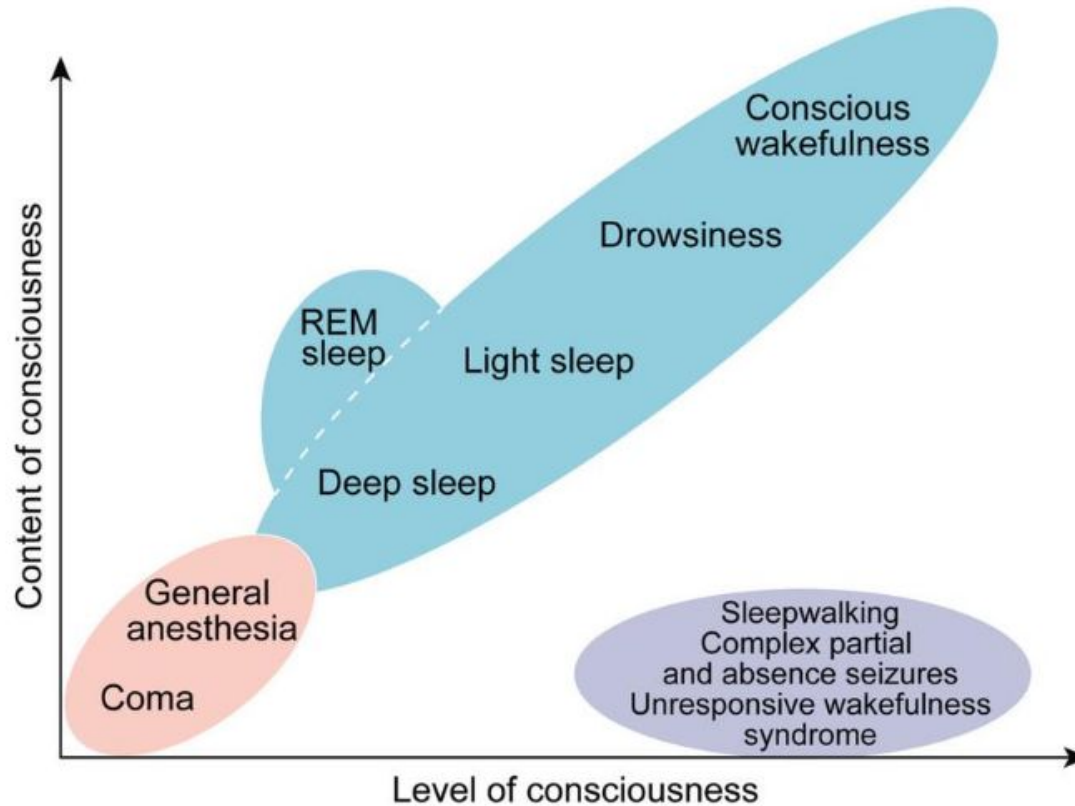
By "consciousness" I mean those states of sentience or awareness that typically begin when we wake up in the morning from a dreamless sleep and continue throughout the day until we fall asleep again. Other ways in which consciousness can cease is if we die, go into a coma, or otherwise become unconscious. (John Searle, 1993)

Bi-dimensional definition of consciousness



Laureys, 2004

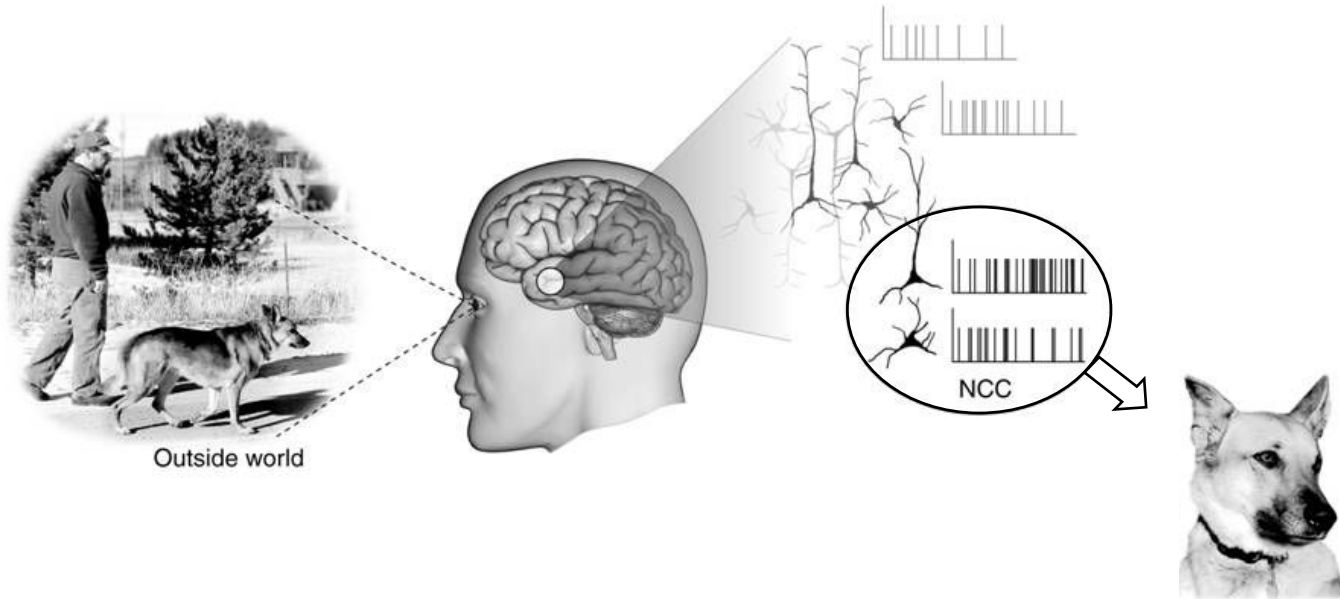
Bi-dimensional definition of consciousness



Laureys, 2004

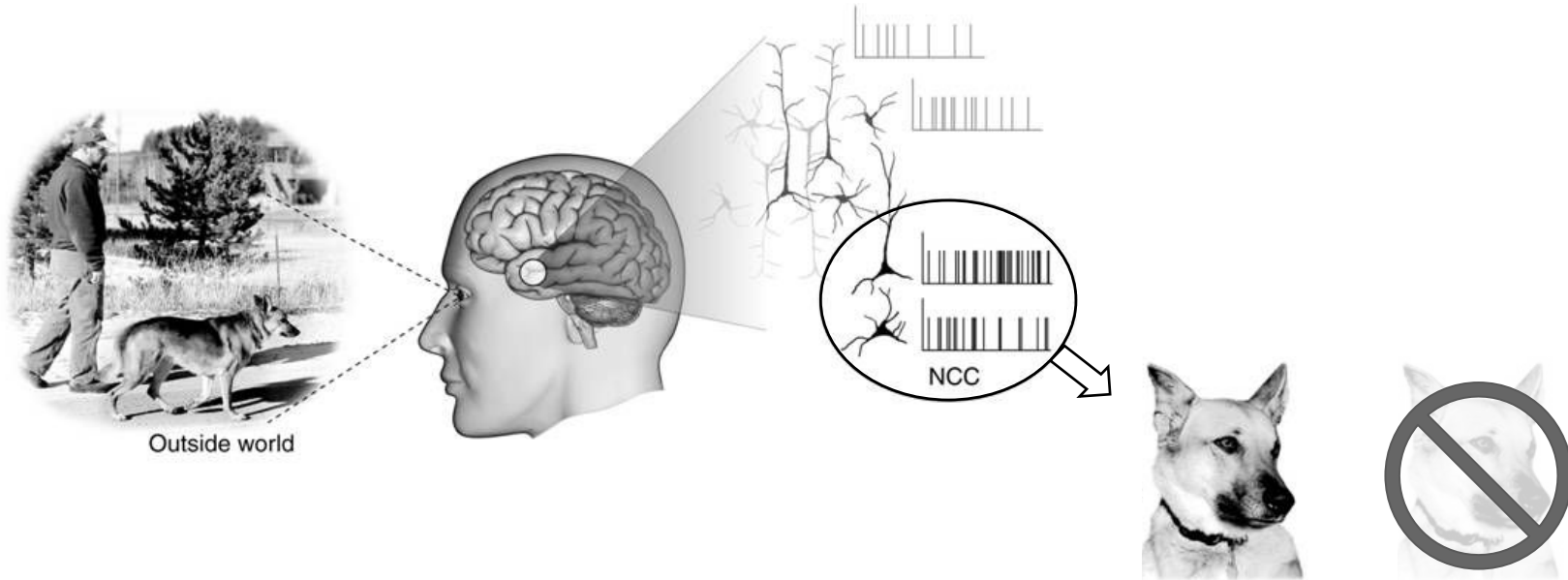
Neural correlates of consciousness (NCC)

The minimal neuronal mechanisms jointly sufficient for any one specific conscious percept (Crick & Koch, 1990)



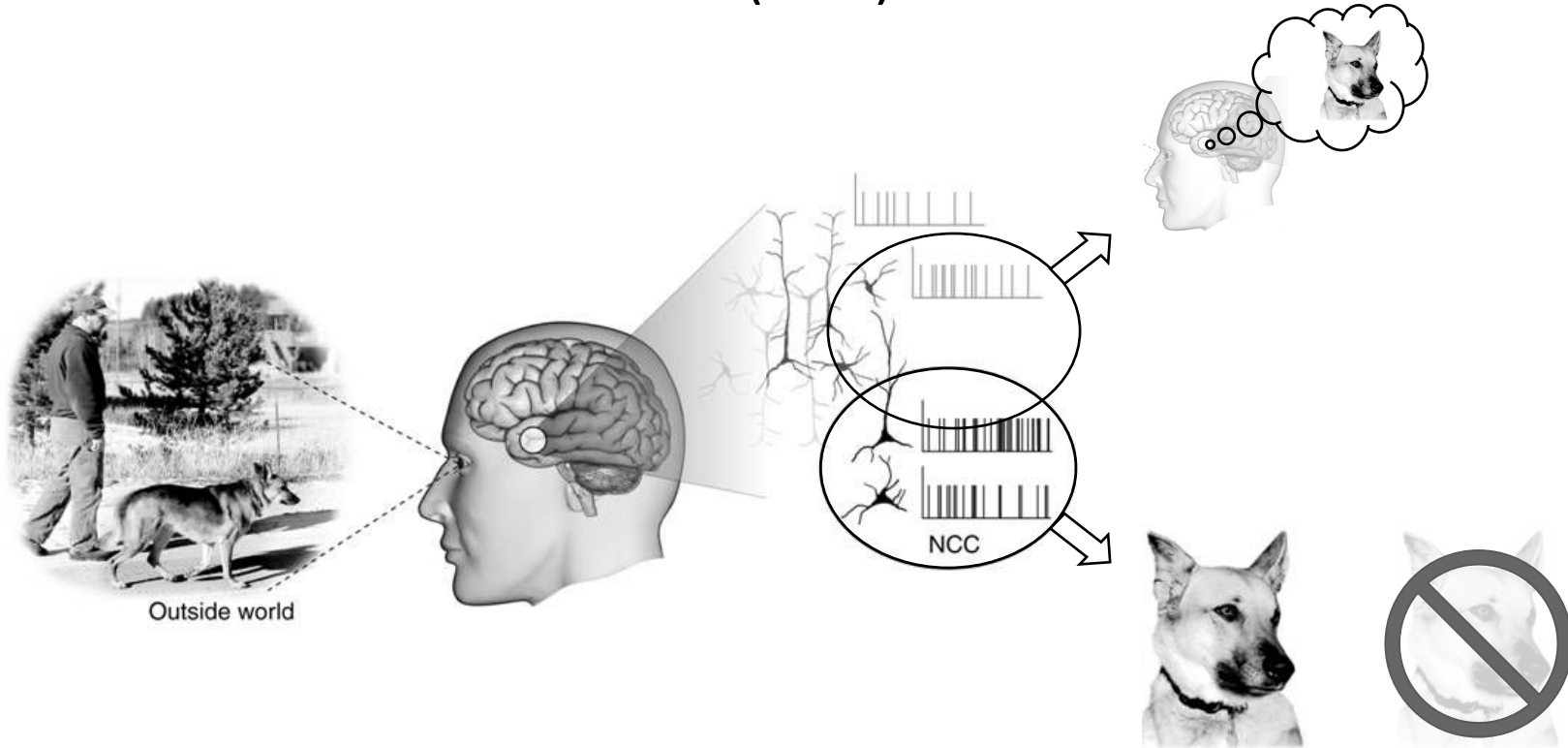
Neural correlates of consciousness (NCC)

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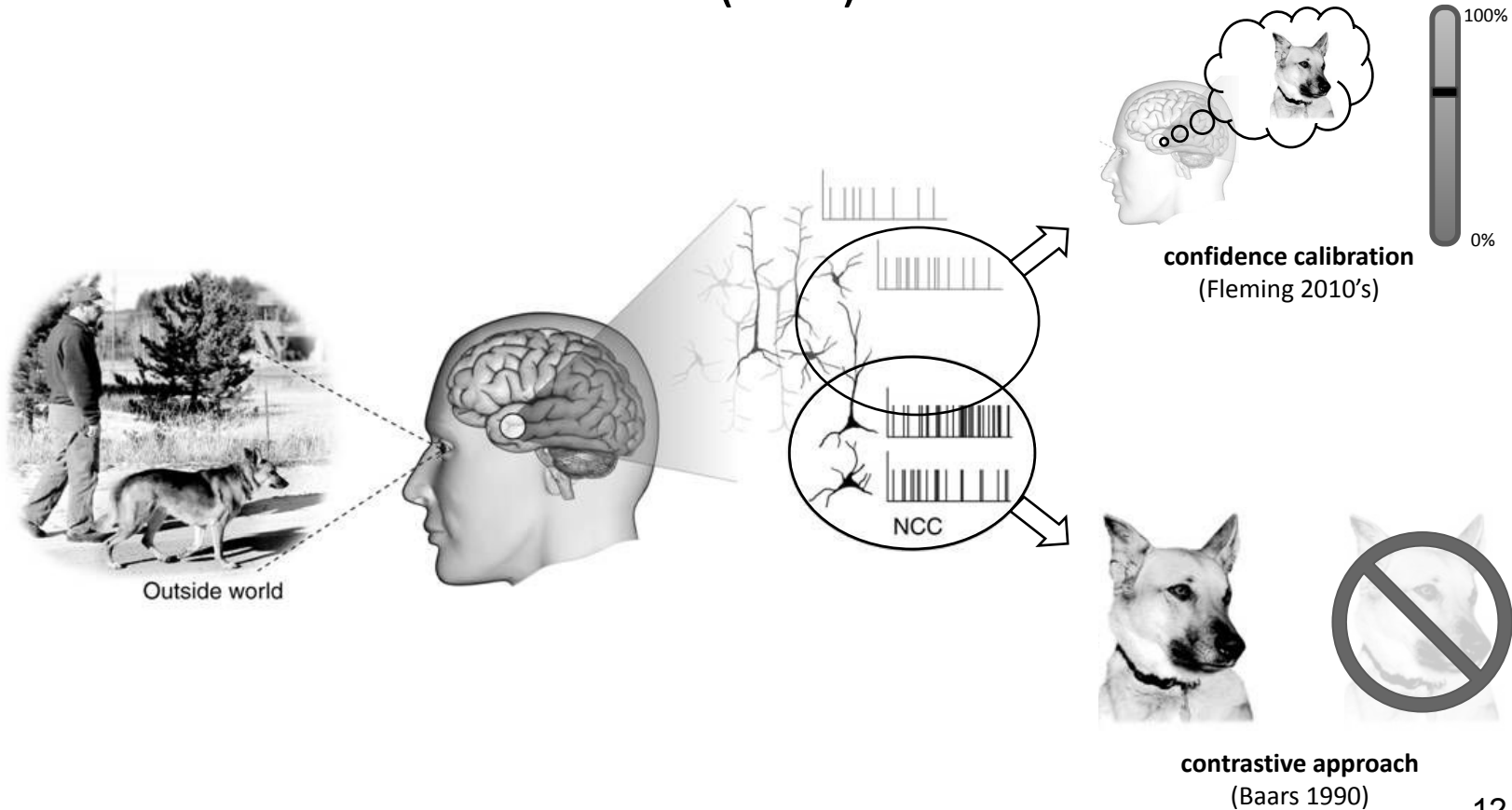
contrastive approach
(Baars 1990)

Neural correlates of consciousness (NCC)



contrastive approach
(Baars 1990)

Neural correlates of consciousness (NCC)



The NCC wager

*I bet you a case of wine
that NCCs will be clear
within 25 years!!*



Christof Koch

ASSC 2, Bremen, 1998

... No it won't!

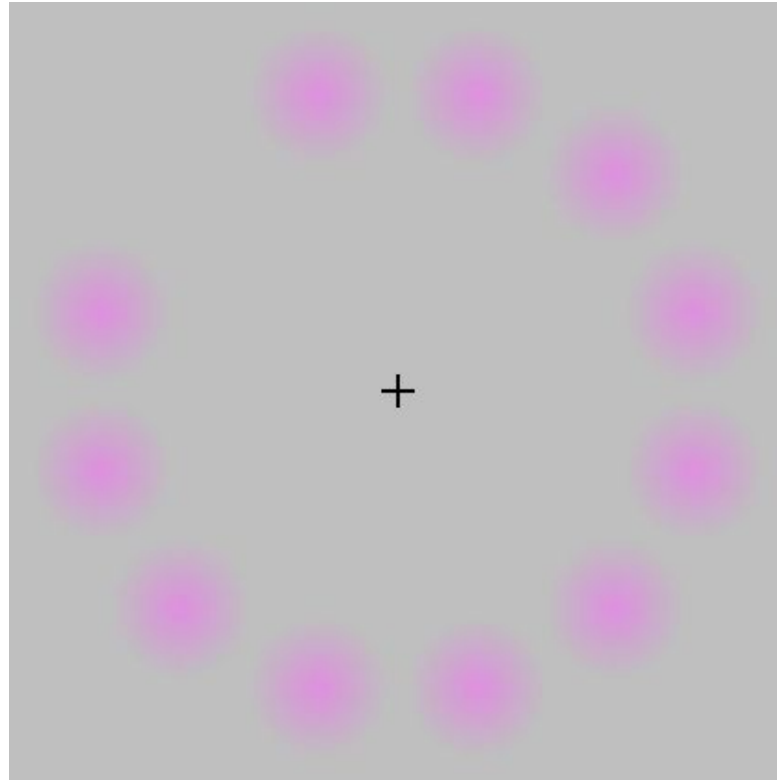


David Chalmers



ASSC 26, NYC, 2023

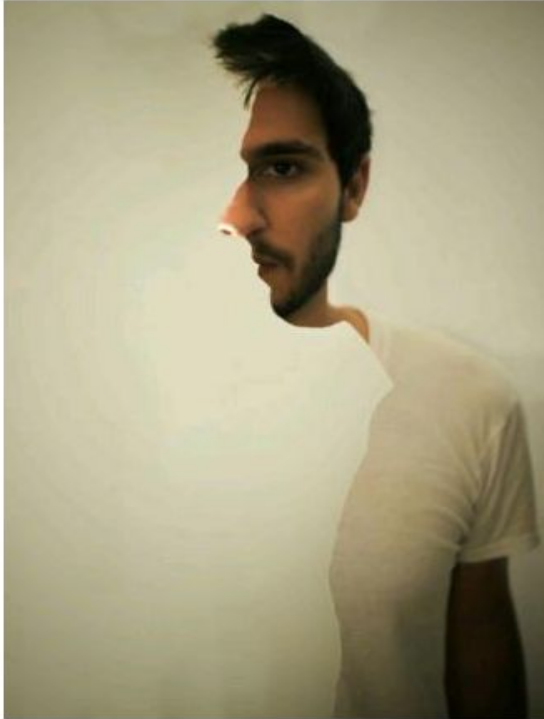
Illusions and conscious contents



Lilac chaser illusion (Hinton, 2005)

Illusions and conscious contents

Bistable stimuli



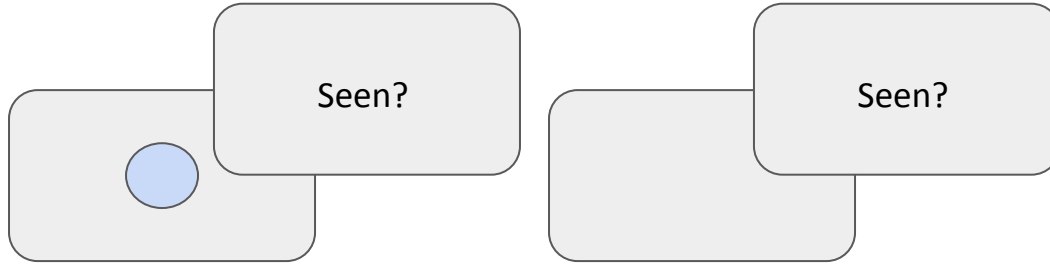
Blake & Logothetis, 2002

Binocular rivalry

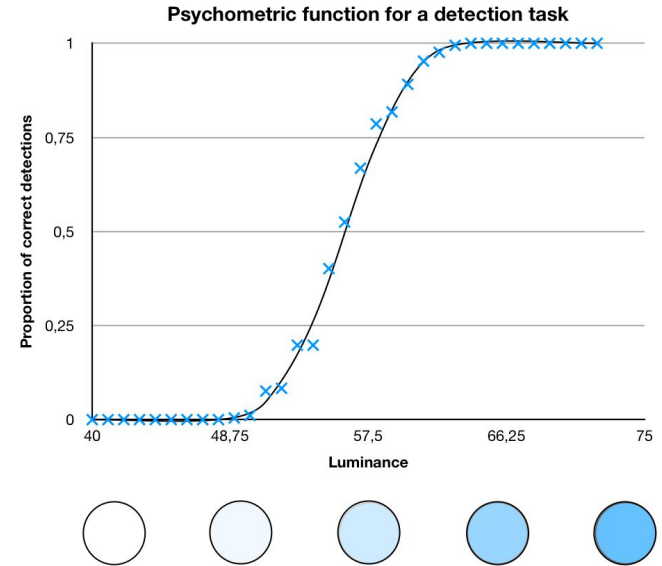


Tong et al., 1998

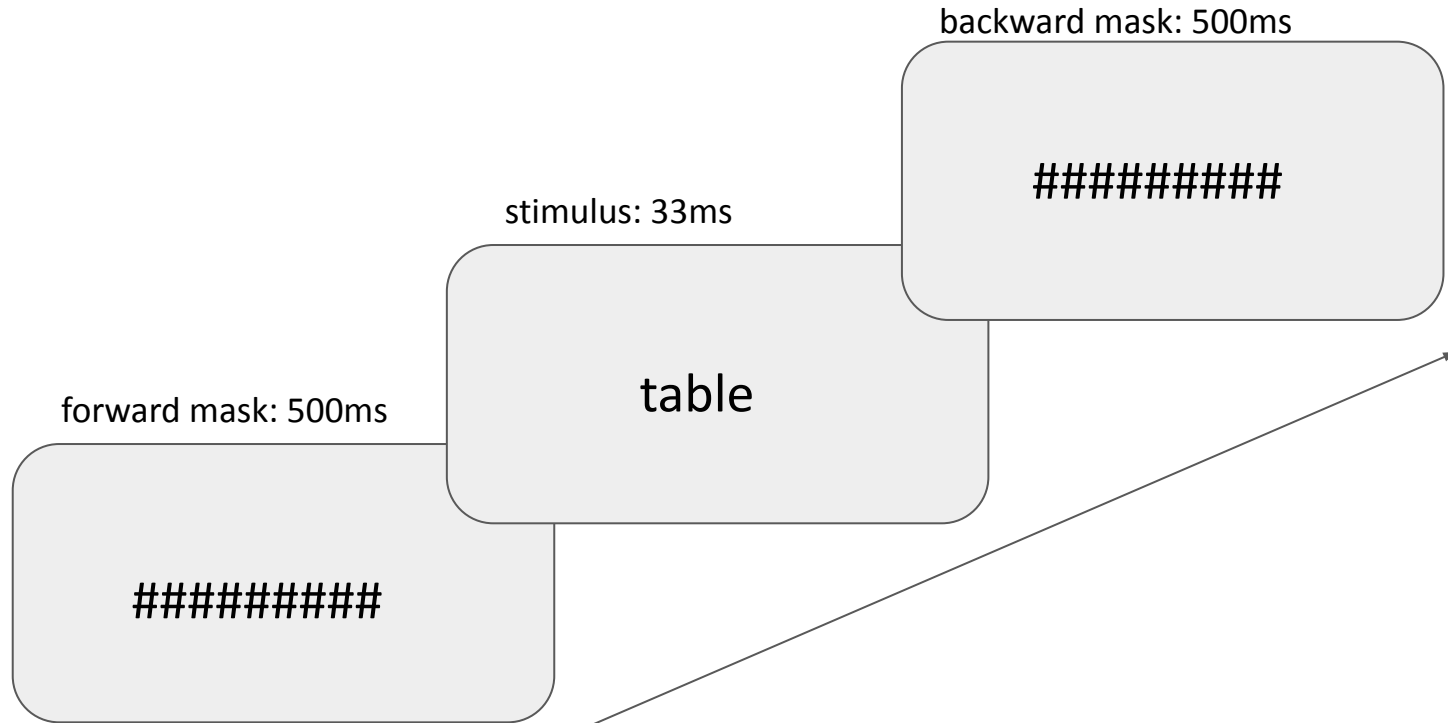
Detection and conscious contents

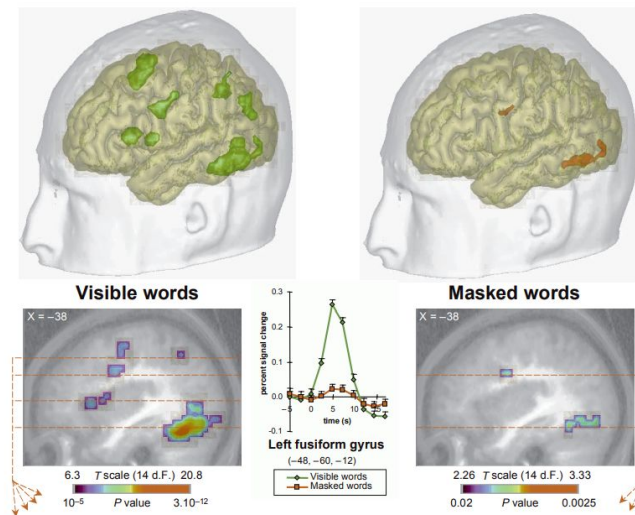
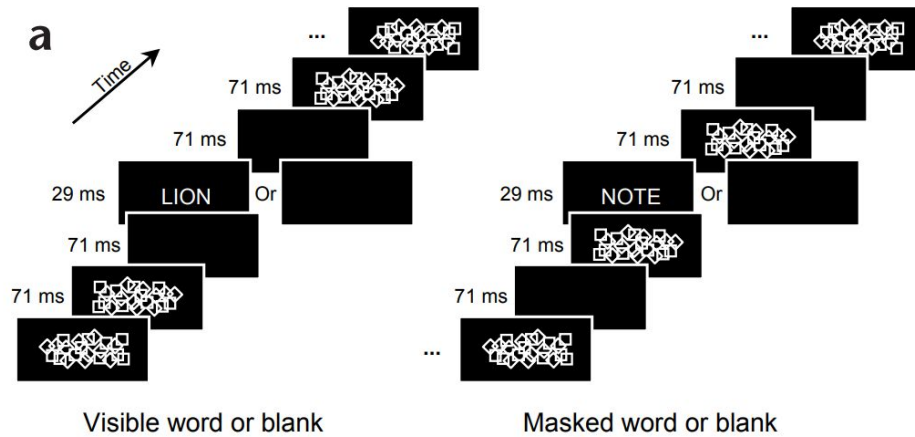


		Response	
		YES	NO
Stimulus	YES	Hit	Miss
	NO	F.A.	C.Rej

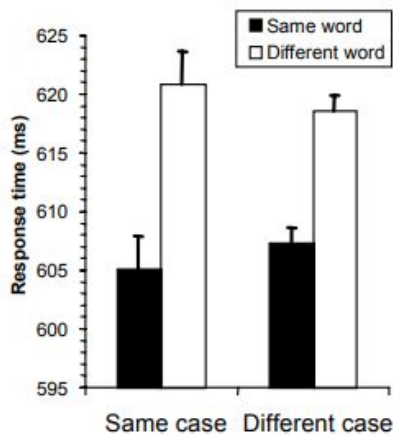
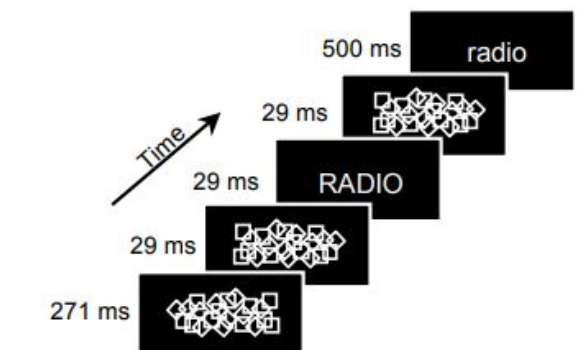


Visual Masking

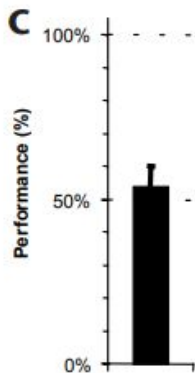




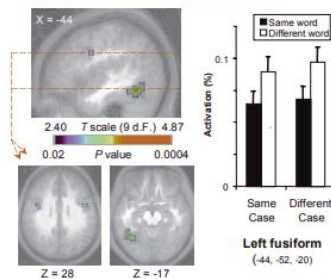
Dehaene et al., 2001



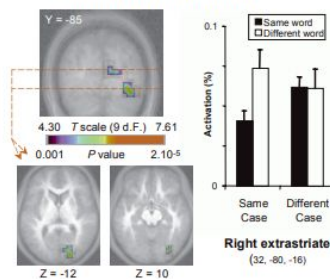
Dehaene et al., 2001



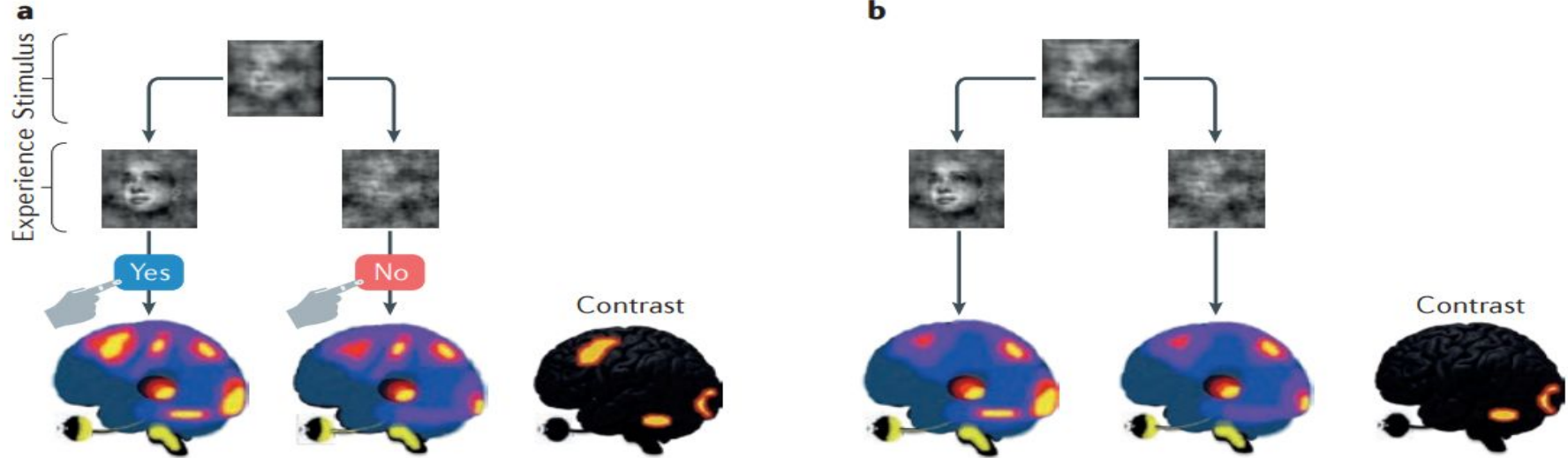
Case-independent priming



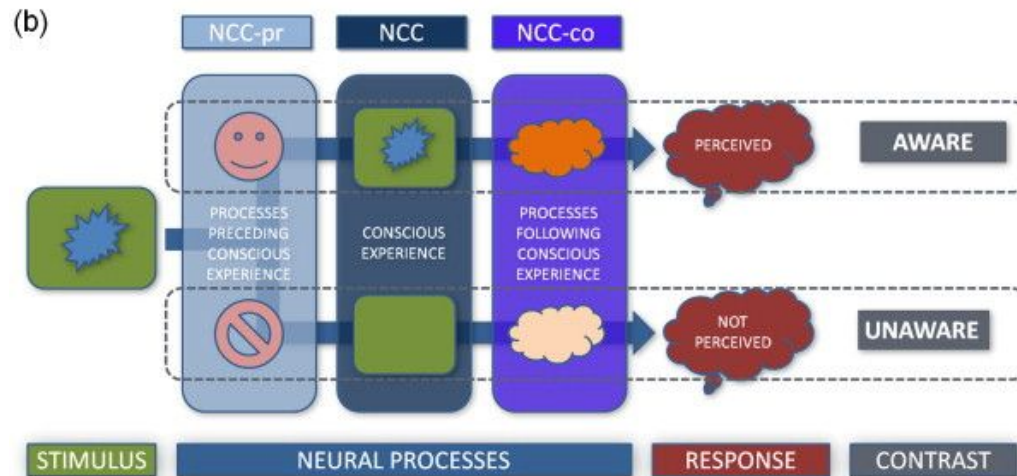
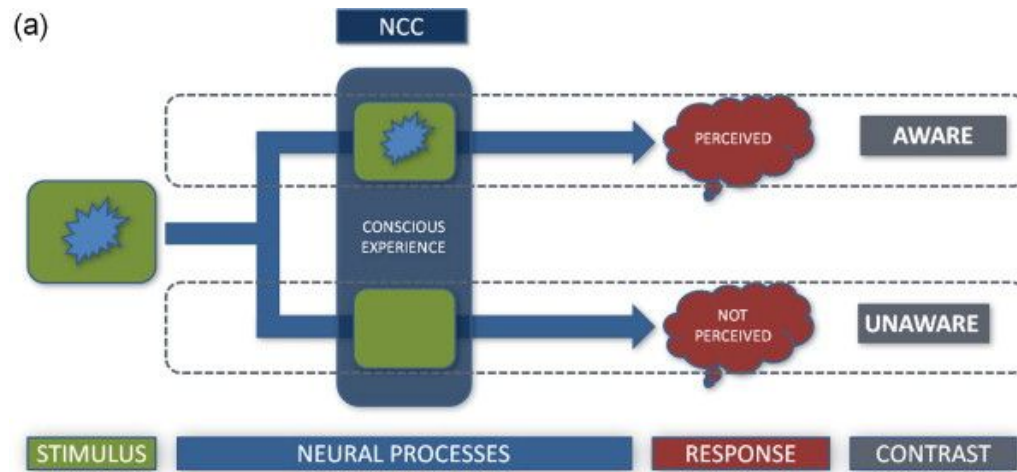
Case-specific priming



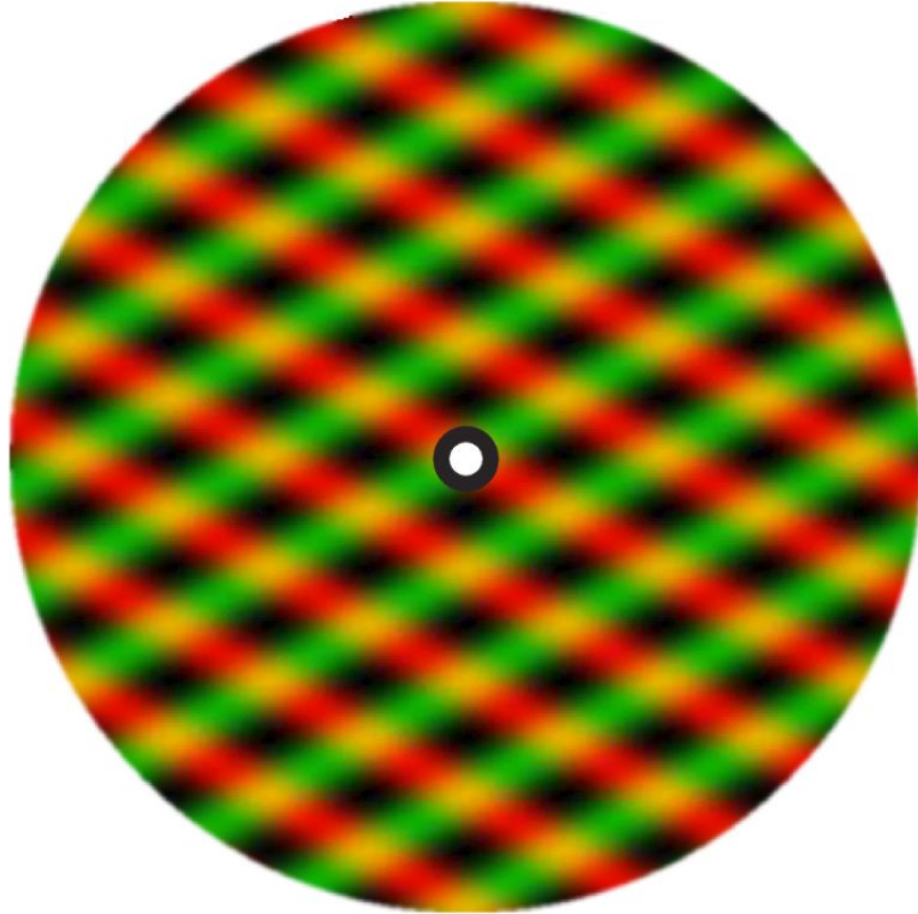
Confounding factors in NCCs



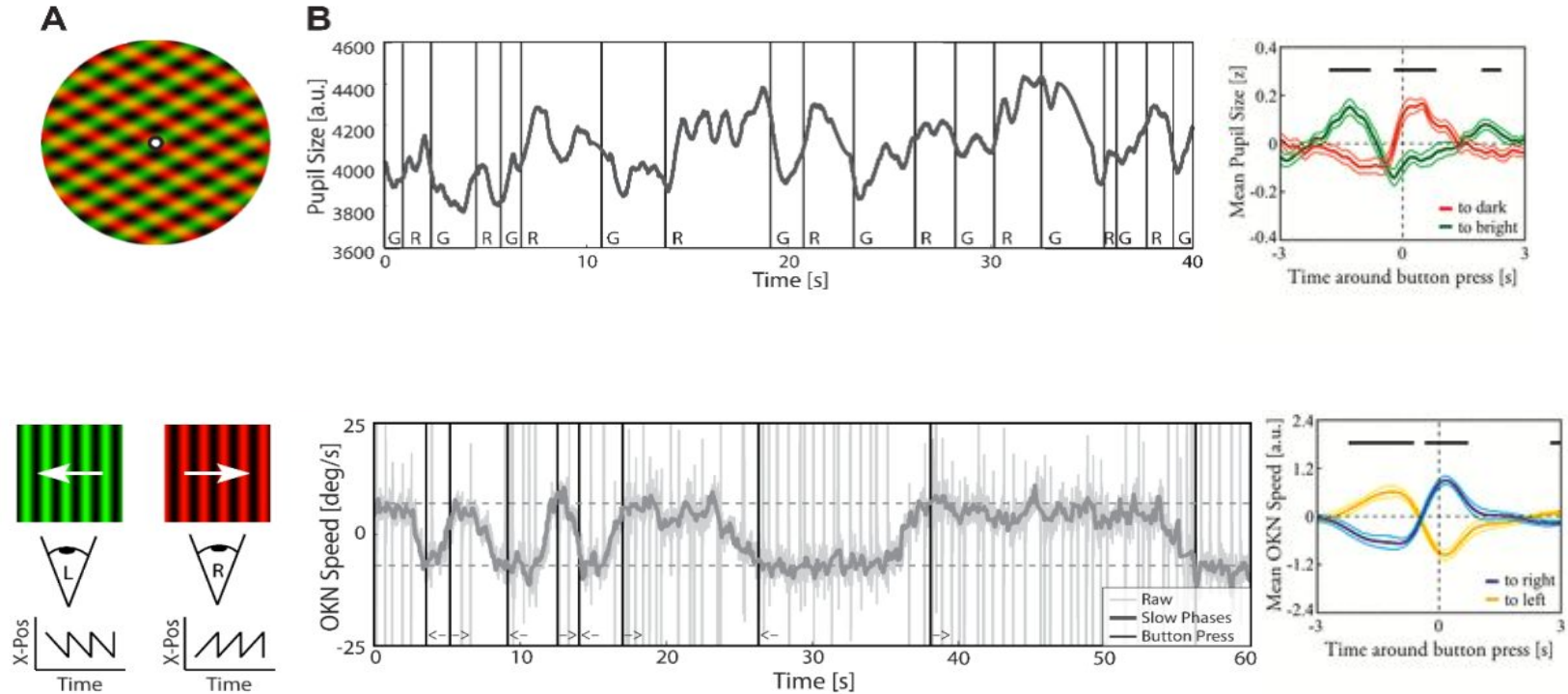
Koch et al., 2016



Inferring conscious contents

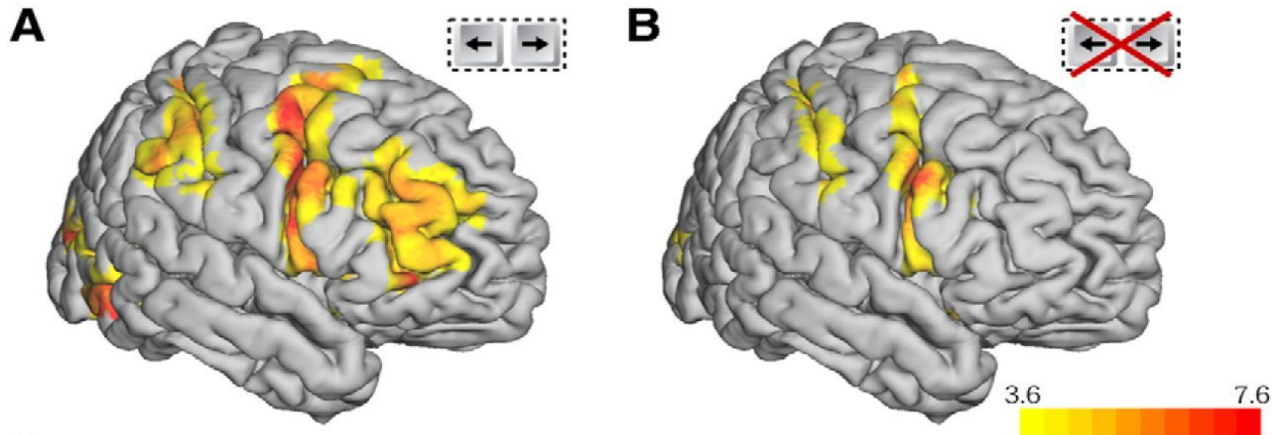


Inferring conscious contents

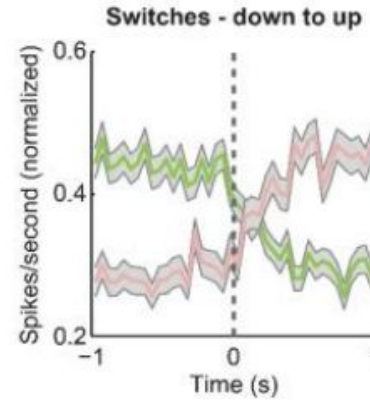
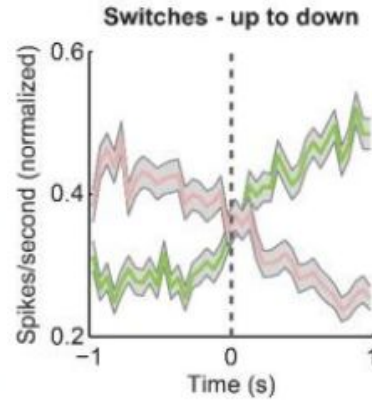
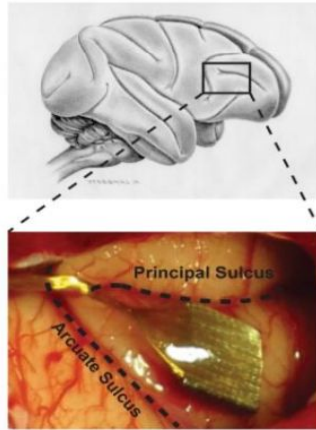


Frässle et al., 2014

Inferring conscious contents



Inferring conscious contents



Kapoor et al., 2020

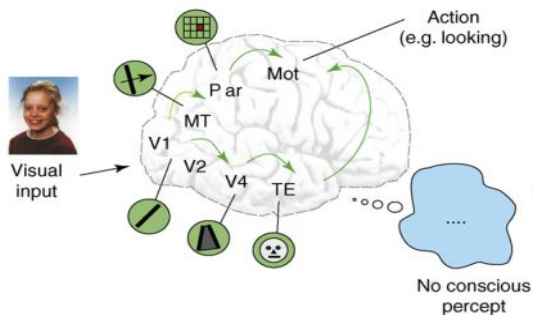
Theories of consciousness: an overview

- Disclaimer: we only cover a small subset of neuroscientific theories
(*no quantum physics, dualism, spiritualism, etc.*)
- No consensus as the field is relatively young
- Goals for a theory of consciousness:
 - explain empirical data (why cerebellum not involved, why fades out at night, etc.)
 - make predictions (x is conscious, y is unconscious)
 - falsifiable (if data shows x, then T is false)
- Easy problem: how a stimulus becomes reportable
Hard problem: how and why physical processes give rise to experience
- **Selected candidates**
 - Recurrent Processing Theory (RPT)
 - Global Neuronal Workspace (GNW)
 - Higher Order Theory (HOT)
 - Integrated Information Theory (IIT)

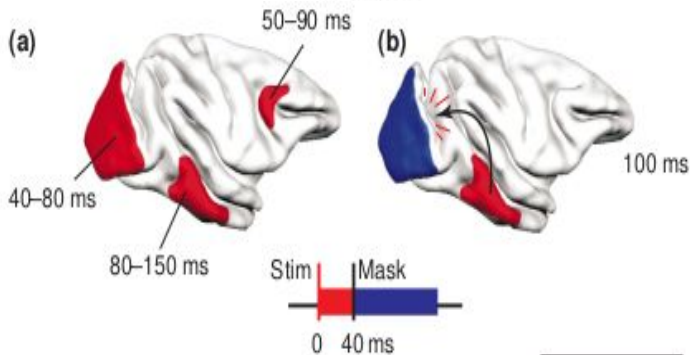
Recurrent Processing Theory

Visual consciousness emerges when recurrent neural activity allows different specialized visual regions to communicate.

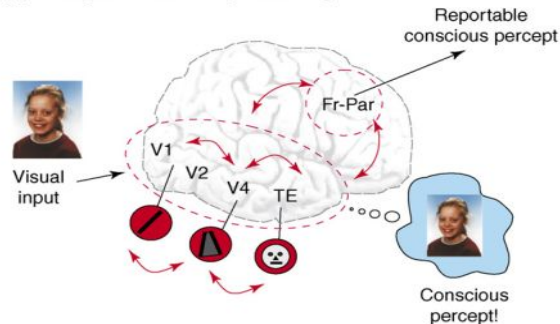
(a) The feedforward sweep



(b) Localized recurrent processing

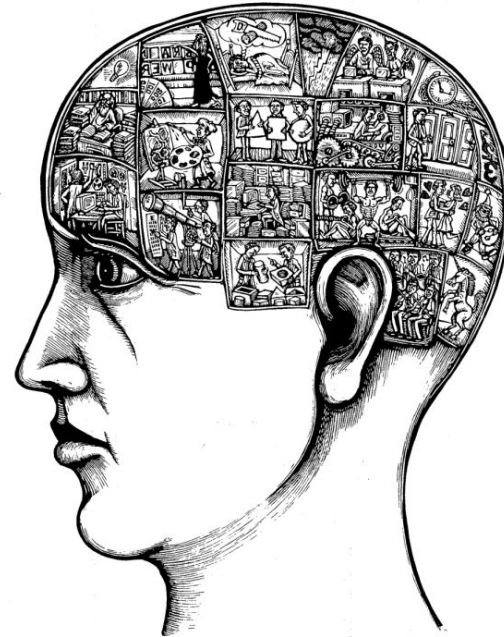
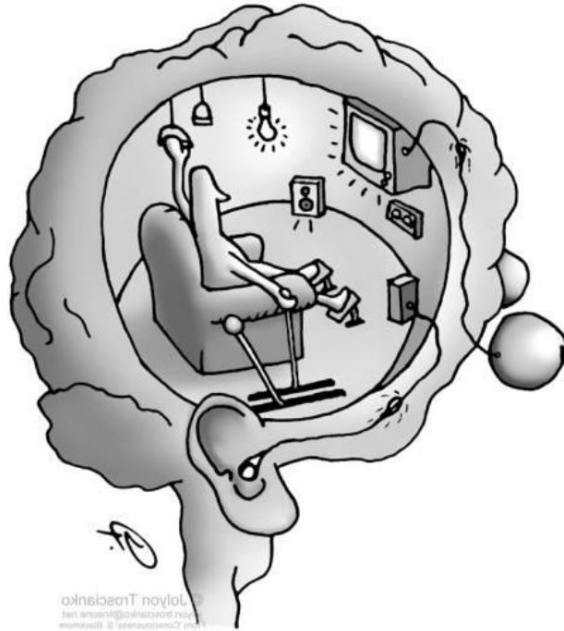


(c) Widespread recurrent processing



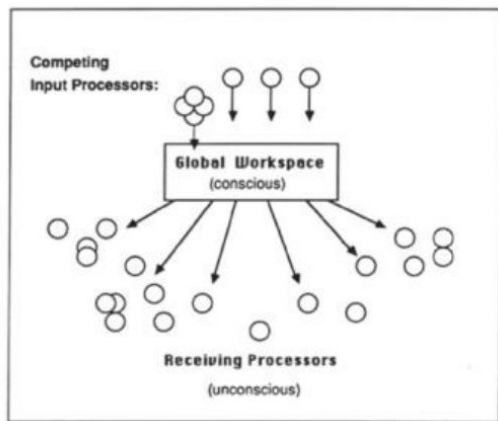
Lamme, 2006

Global Neuronal Workspace

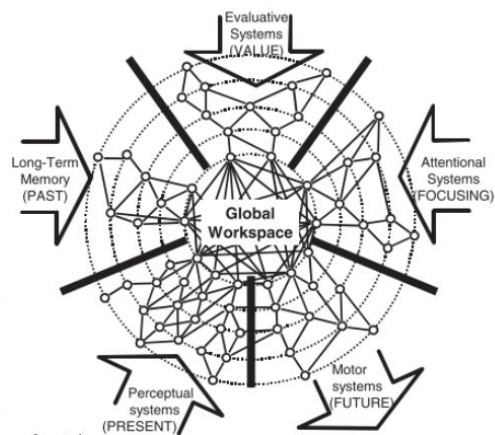


Global Neuronal Workspace

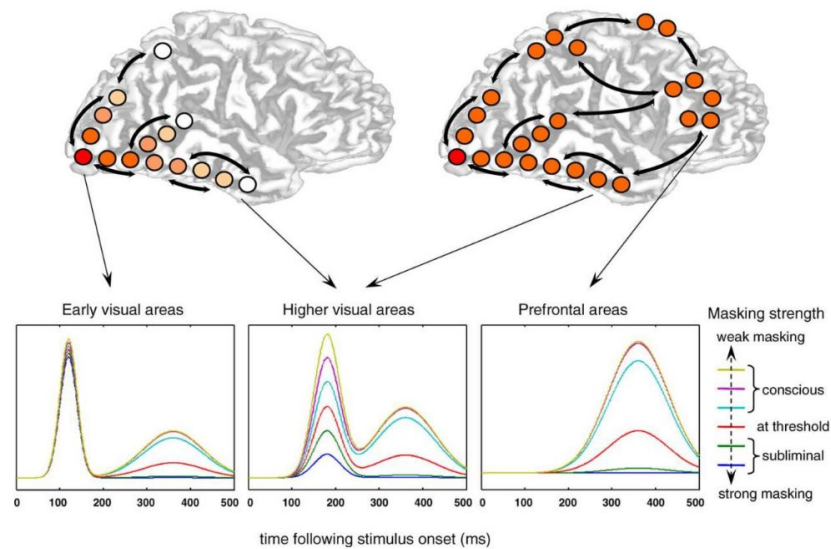
Consciousness is instantiated by the global broadcasting and amplification of information across an interconnected network of prefrontal-parietal areas and many high-level sensory cortical areas.



Baars, 1988

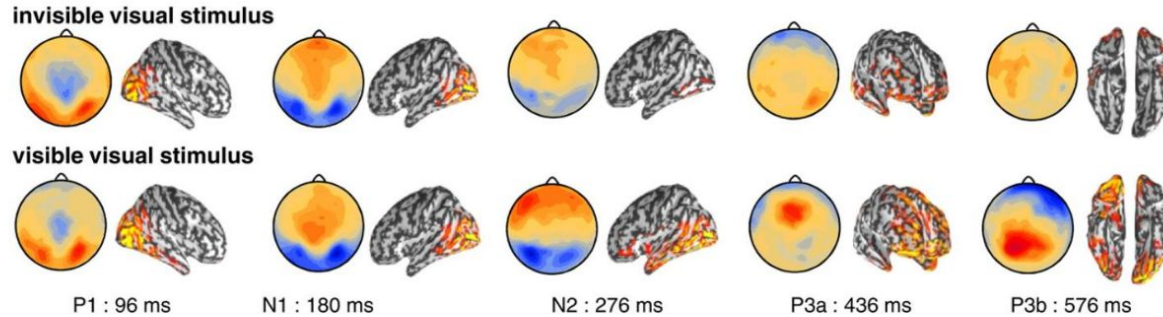


Dehaene et al, 1998

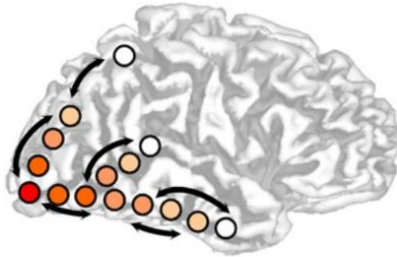


Del Cul et al, 2006

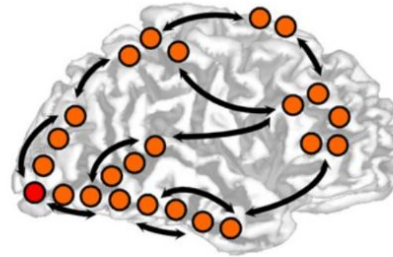
Global Neuronal Workspace



**Feed-forward propagation
(subliminal processing)**



**Reverberating global neuronal workspace
(conscious access)**

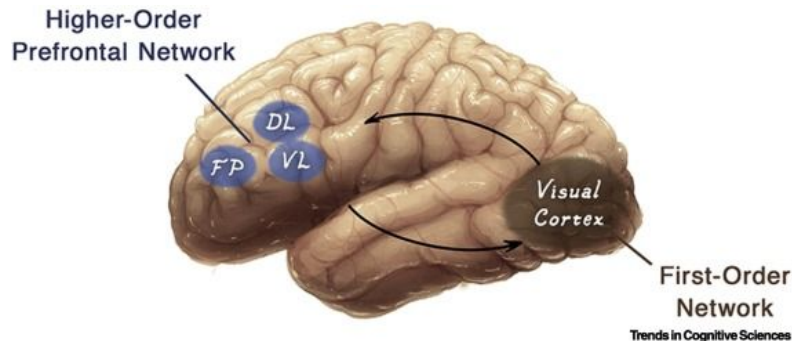
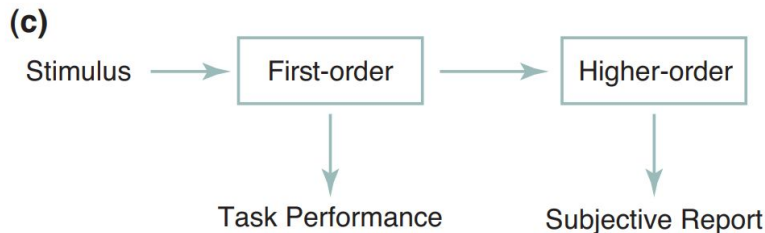


Consciousness *is* global broadcasting of information

Higher Order Theory

1. *A first-order representation is not enough (i.e., not representational)*
2. *To be conscious, an organism must be aware of itself as being in a certain first-order state (transitivity principle: monitoring/meta-representation)*

=> *Consciousness crucially depends on higher-order mental representations that represent oneself as being in particular mental states: A mental state X is conscious if, and only if, one has a higher-order representation to the effect that one is currently representing X.*



Integrated Information Theory

Consciousness reflects the amount of maximally irreducible integrated information generated by certain neuronal architectures

Intrinsic existence



Composition



Information



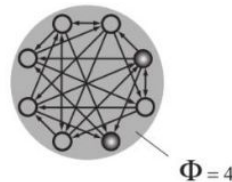
Integration



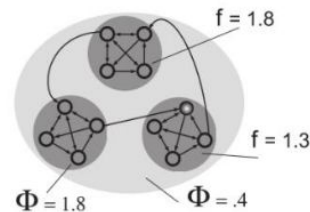
Exclusion



A CORTICOTHALAMIC SYSTEM

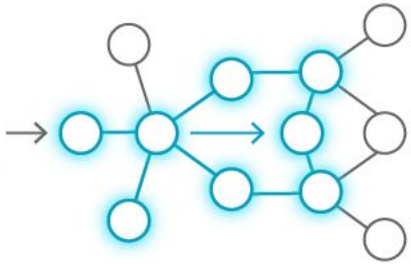


B CEREBELLAR SYSTEM



Summary: premises

Recurrent Processing Theory



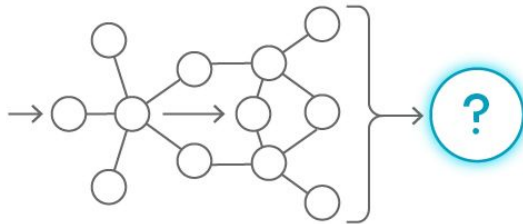
Electrophysiology: feedback

Global Workspace Theory



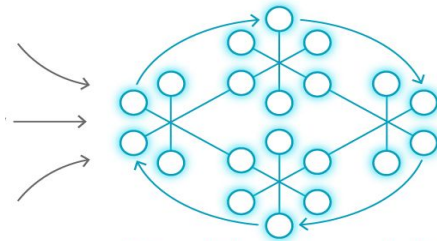
Cognitive Sciences: broadcast

Higher Order Theory



Philosophy: representation

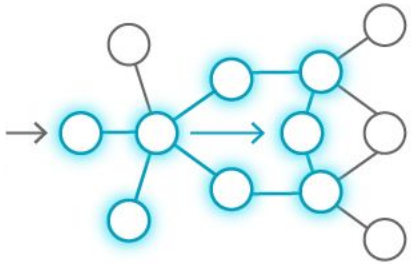
Integrated Information Theory



Phenomenology: integration

Summary: functions

Recurrent Processing Theory



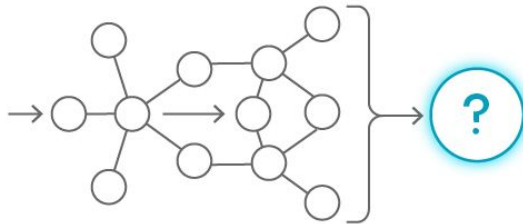
Perceptual organisation

Global Workspace Theory



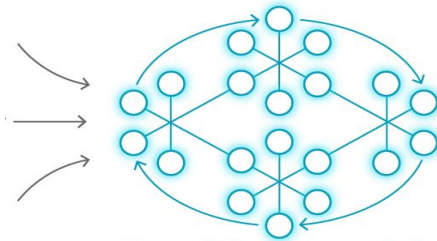
Integration / Flexibility / ...

Higher Order Theory



None / Reality monitoring

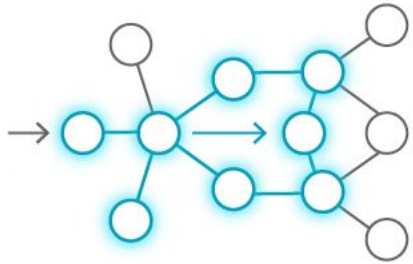
Integrated Information Theory



None

Summary: anatomy

Recurrent Processing Theory



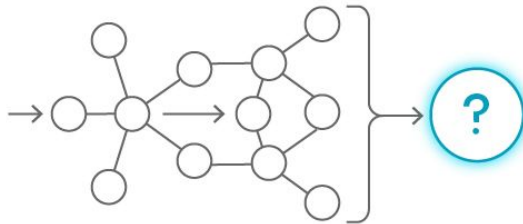
Sensory cortices

Global Workspace Theory



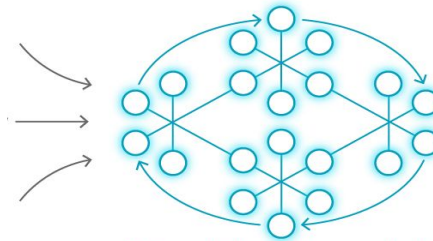
Frontoparietal cortex

Higher Order Theory



Prefrontal cortex

Integrated Information Theory



Posterior hotzone

Summary: problems

Formal criticisms

- Small network & panpsychism
- Unfolding argument
- Generalizability to other systems

Common pitfalls

- Distinct levels of explanation and unclear predictions
- Most empirical studies report positive evidence (post-hoc?)
- Need for more diagnostic empirical evidence



The Integrated Information Theory of Consciousness as Pseudoscience

AUTHORS

IIT-Concerned, [Stephen Fleming](#), Chris Frith, Mel Goodale, [Hakwan Lau](#), [Joseph E LeDoux](#), [Alan L. F. Lee](#), [Matthias Michel](#), Adrian Owen, [Megan A. K. Peters](#), [Heleen A Slagter](#)

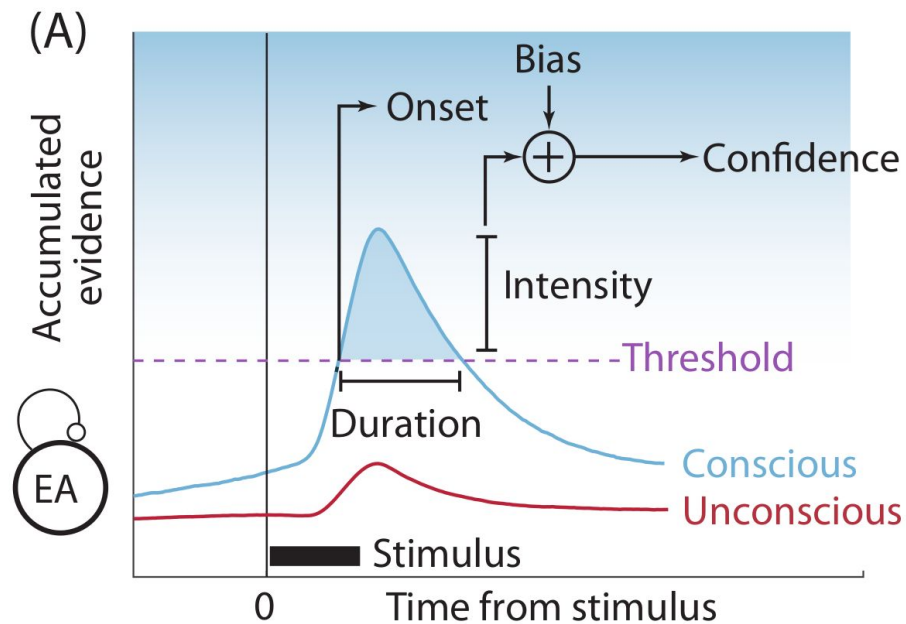
AUTHOR ASSERTIONS

Conflict of Interest: No ▼

Public Data: Not applicable ▼

Preregistration: Not applicable ▼

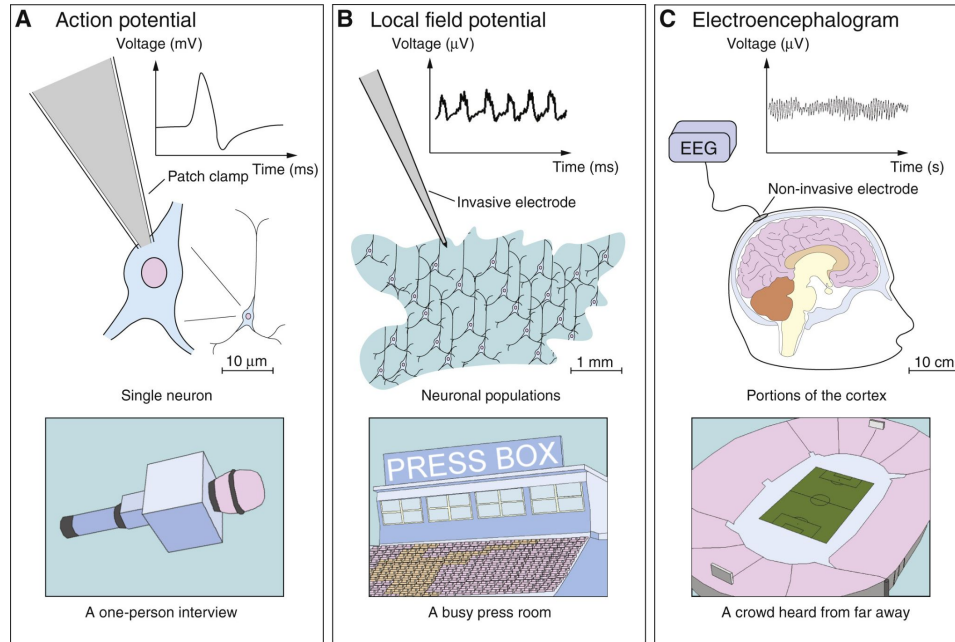
Evidence accumulation and consciousness



Pereira, Perrin, & Faivre, TICS 2022

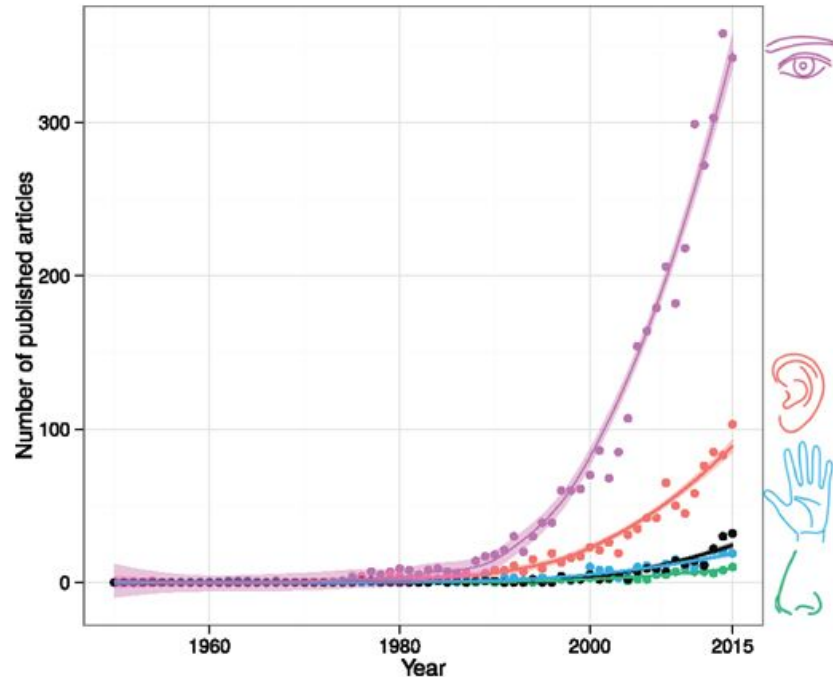
Take home message

- NCCs described at different scales: trade-off resolution / coverage



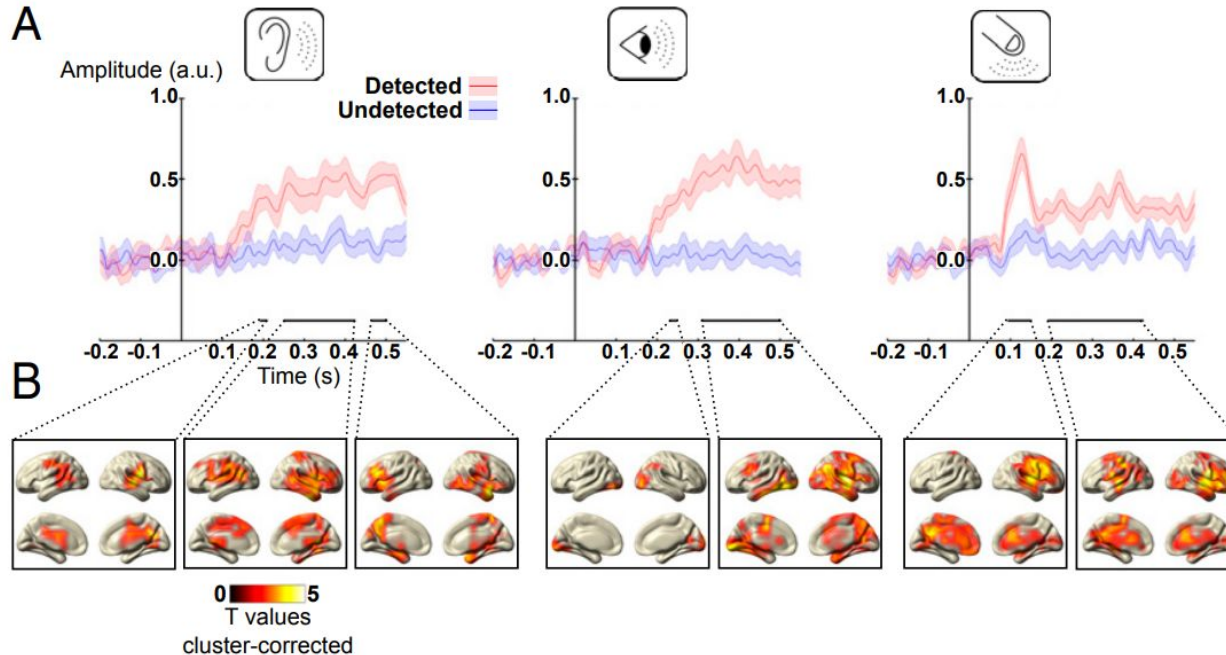
Take home message

- Consciousness beyond vision



Take home message

- Consciousness beyond vision



Take home message

- Important to distinguish prerequisites and consequences



Take home message

- Central to important ethical issues

